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Département Of Industrial and Mechanical Engineering

Subject: Organ of Mechanic

Topic: Coupling

Lecture : **M. Mut Mesa (Couse et TD)**

Group: I4-GIM-A1

Student:	**ID**
1. BAN THUNPISEY	e20220860
2. BUNTHEN CHANSOYUTHARITH	e20220219
3. CHAO SIVCHHUONG	e20220409
4. CHAT CHAN PARADY	e20221655
5. CHEA SENGTRY	e20220606
6. CHEN CHANRAKSMEY	e20220632

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Coupling

I. Introduction

1. Definition

In mechanical power transmission systems, a **coupling** is an essential component used to connect two rotating shafts together so that power can be transmitted from one shaft to another. Couplings are widely used in machines because most mechanical systems are built with multiple shafts that must rotate together smoothly and safely. Without couplings, machines would experience misalignment, high vibration, and rapid wear of parts.

A **coupling** is a device used to connect two shafts together at their ends for the purpose of transmitting power. The primary purpose of couplings is to join two pieces of rotating equipment while permitting some degree of misalignment or end movement or both. In a more general context, a coupling can also be a mechanical device that serves to connect the ends of adjacent parts or objects.

Couplings do not normally allow disconnection of shafts during operation, however there are torque limiting couplings which can slip or disconnect when some torque limit is exceeded. Selection, installation and maintenance of couplings can lead to reduced maintenance time and maintenance cost.

2. Objective of Using Coupling

Figure 1: The coupling

The goals of using coupling:

- Transmit power and torque between two shafts
- Accommodate shaft misalignment
- Absorb shock loads and vibrations
- Reduce wear on shafts and bearings
- Protect mechanical components from overload
- Improve efficiency and reliability of power transmission

3. Scope of works

- **Team responsibilities**

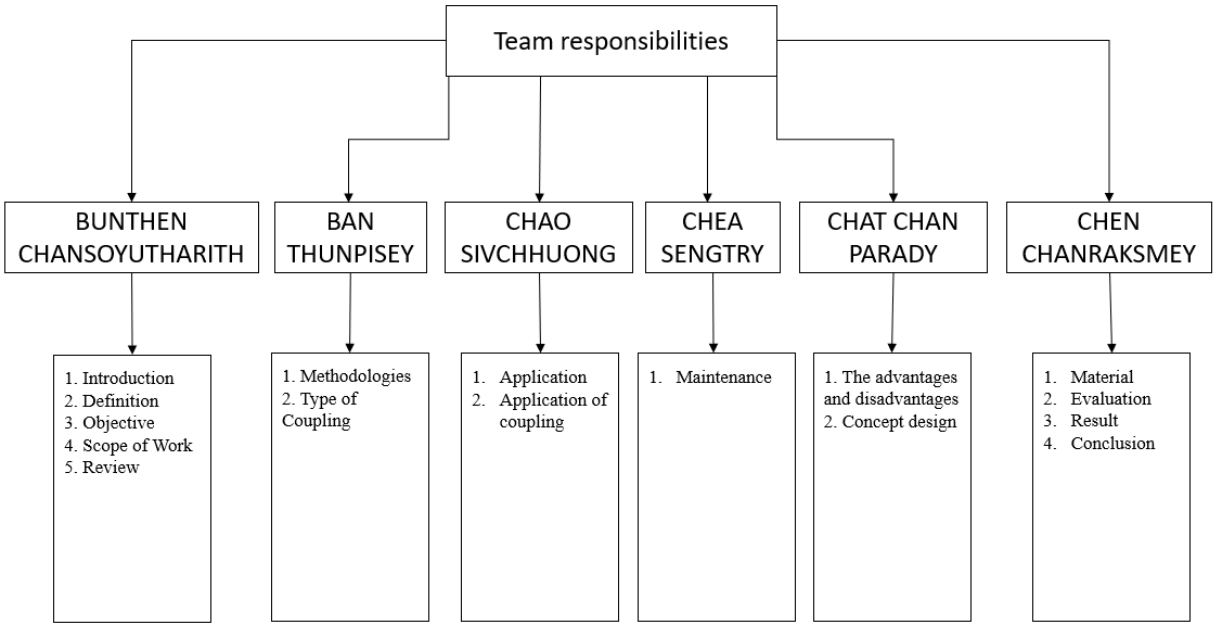


Figure 3:Team responsibilities

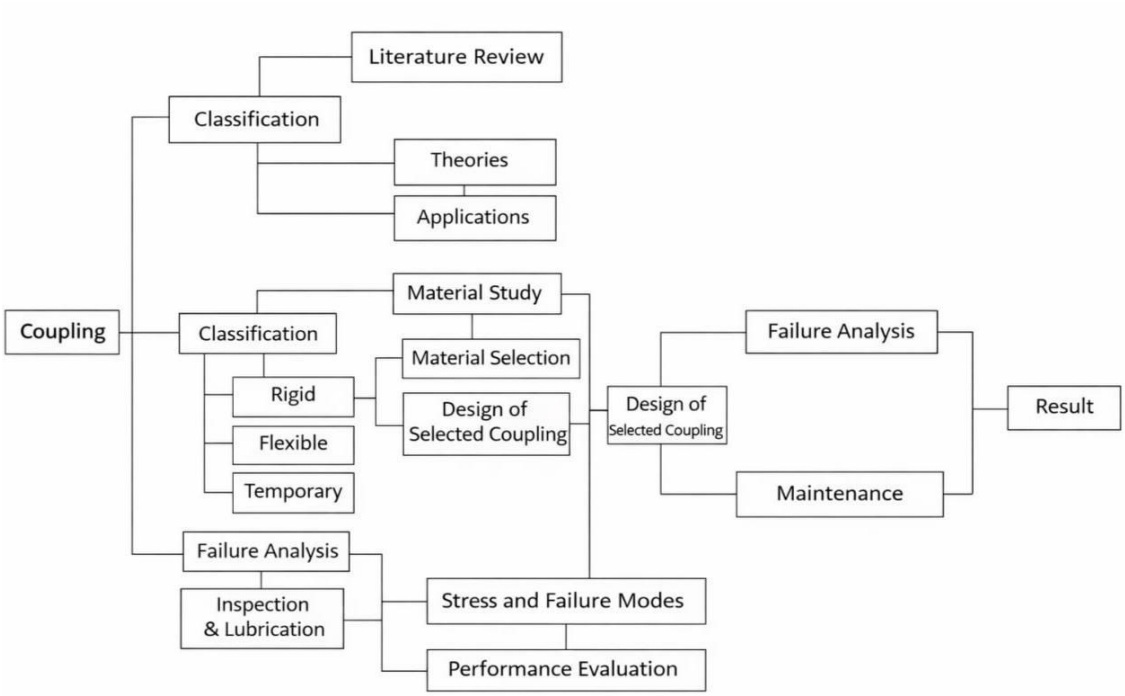


Figure 2: Flow of work

II. Review

Based on previous studies and engineering practices, couplings play an important role in mechanical power transmission systems. The main review points are listed below:

- Couplings are used to connect two rotating shafts to transmit torque and motion.
- They help accommodate shaft misalignment such as angular, parallel, and axial misalignment.
- Couplings reduce vibration and absorb shock loads during machine operation.
- Different types of couplings are used depending on working conditions:
    - Rigid couplings for accurately aligned shafts
    - Flexible couplings for vibration reduction and misalignment compensation
    - Universal and safety couplings for angular movement and overload protection
- Material selection affects coupling strength, durability, and wear resistance.
- Proper installation and regular maintenance improve coupling performance and service life.

III. Methodologies

1. Types of coupling

Couplings are primarily classified based on how they connect and disconnect shafts: Permanent Couplings and Temporary Couplings.

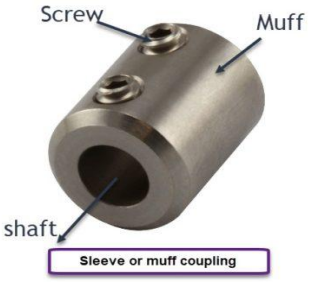
- **Permanent Couplings**

Permanent couplings create a fixed joint between pipes or shafts. They provide a strong, rigid connection for power transmission or fluid flow in static systems and are not intended to be disconnected during operation.

- **Rigid Couplings**

Rigid couplings connect shafts that must be perfectly aligned. They provide a solid, high-precision connection and offer maximum torsional stiffness. In the rigid coupling, it has many types like:

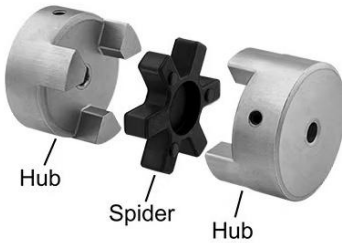
❖ **Sleeve or Muff Coupling:** Used when two shafts are perfectly aligned. They transmit low to medium torque, often utilizing a hollow cylinder and a keyway.



- **Flange Coupling:** Widely used to connect shafts of the same or different diameters. These helps maintain alignment and are capable of supporting heavy loads. They are commonly utilized in pressurized piping systems.

- **Clamp or Split-Muff Coupling:** Used to connect shafts of the same diameter. These are suitable for heavy-duty and moderate-speed applications because they can be installed or removed without shifting the shafts.

- **Flexible Couplings**
  - **Jaw Coupling:** Well-suited for high-speed rotation and high-precision transmission. While they are common, their ability to handle extreme misalignments is limited.



  - **Gear Coupling:** Transmits torque through meshing gears. Their design accommodates significant axial and angular misalignments, making them ideal for high-power, heavy-duty industrial systems.

- **Bellows Coupling:** Uses the flexibility of a thin-walled metal bellow to compensate for misalignment. They offer excellent resistance to high temperatures, corrosion, and vacuum conditions.

- **Universal Coupling (Hooke's Joint):** Connects two rotating shafts whose axes intersect at an angle, allowing power transmission even as the angle changes.

- **Homokinetic (CV) Joints:** A mechanical coupling that transmits power between shafts at varying angles while maintaining a constant rotational speed without vibration.

- **Temporary Couplings (Clutches)**

  Temporary couplings allow shafts to be easily and intentionally connected or disconnected on demand, often while the machinery is in motion.

- **Jaw Clutch:** Uses interlocking jaws or teeth to engage the shafts. This usually requires the shafts to be stationary or moving at very low speeds for engagement.

- **Fluid or Hydraulic Coupling:** Transmits torque using fluid momentum, providing a "soft start" with reduced shock. While technically always connected, it allows for slippage at low speeds, effectively decoupling the shafts.

- **Torque-Limiting (Safety) Clutches:** Designed to automatically slip or disconnect when the transmitted torque exceeds a predetermined limit, protecting machinery from overload damage.

2. The material of coupling

Structure-activity collective properties underlying self-assembled superstructures. Strong electronic coupling mechanism and related collective properties. The two most common physical phenomena in strong coupling systems are the Mott insulator-metal transition (MIT) and the Anderson metal-insulator transition, both of which are influenced by the interparticle spacing and degree of order. Currently, research concerning strong coupling materials has focused mainly on **metal chalcogenide QDs.** Compared with bulk materials prepared by traditional methods, a **QD** artificial solid with programmed interparticle spacing and assembled modes can achieve band-like transmission, exhibiting enhanced conductivity and collective electronic properties that validate the mechanisms of MIT and Anderson metal-insulator transition. The MIT depends on the separation between **NPs** [174–176]. With decreasing distance between the surfaces of adjacent metallic spheres δ, wavefunction overlap increases. With further reduction in δ, the conduction gap caused by charging energy ($\varepsilon e=e2/C$) disappears, while the energy band generated by the overlap between **NP** wavefunctions promotes band-like transport and the MIT.

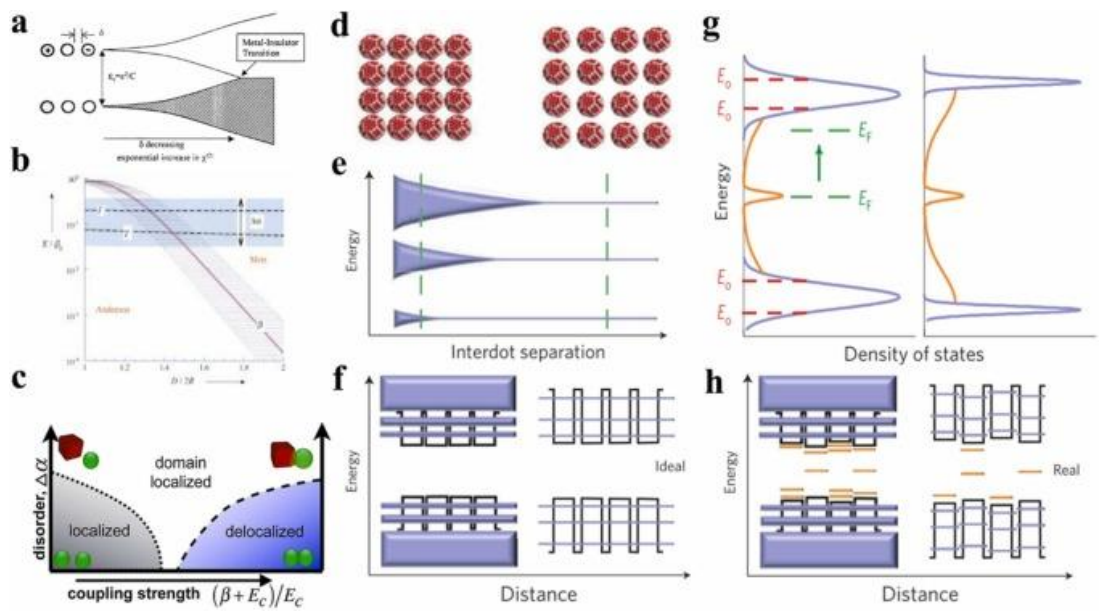


Figure 4: Stress and Strain of material

3. Concept design

3.1. Design Points (Conceptual Intent)

- **Torque Transmission:** Must reliably transfer torque from driver to drive shaft.
- **Misalignment Compensation:** Handle angular, parallel, and axial misalignments within specified limits.
- **Vibration Damping:** Reduce shock loads and oscillations.
- **Overload Protection:** Act as a mechanical safeguard.
- **Ease of Maintenance:** Simple installation, replacement, and inspection.
- **Compactness & Cost:** Small footprint, low manufacturing cost, standard materials.

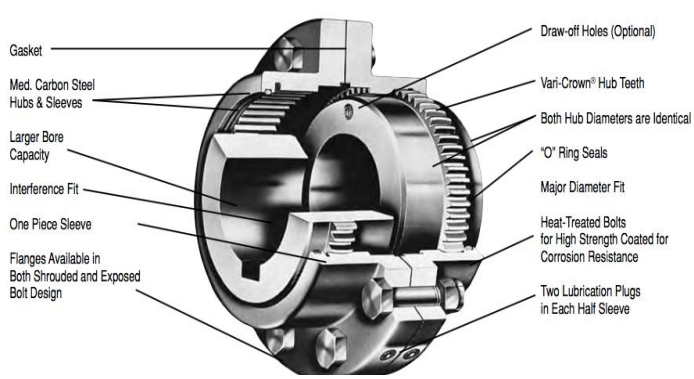


Figure 5: The part's name of coupling

3.2. Governing Formulas

Power Transmission

$$\mathrm{P} = \mathrm{C} * \omega = \{\backslash\mathrm{frac}\ \{\mathrm{C} * \mathrm{N} * \pi\}\{30\}\}$$

$$P = C \cdot \omega = \frac{C \cdot N \cdot \pi i}{30}$$

$P = Powertransmitted\ [W]$

$N = Rotationalspeed[rpm]$

$C = Torque[N \cdot m]$

$\omega = Angularvelocity[rad/s]$

Torque Requirement

$$C = \frac{9550 \cdot P}{N}$$

$C = Torque\ [N \cdot m]$

$P = Power\ [kW]$

$N = Speed[rpm]$

Bolt Diameter Check (for Flange/Plate Coupling)

$$d \geq \sqrt{\frac{8C}{n_b \cdot \pi \cdot D \cdot R_{pg}}}$$

$d = Bolt\ diameter\ [mm]$

$C = Torque\ [N \cdot mm]$

$n_b = Number\ of\ bolts$

$D = Bolt\ pitch\ circle\ diameter\ [mm]$

$Rpg = Allowableshearstress\big((\approx 0.5R_e)\big)$

Shaft Diameter Check

$$d_s \geq \sqrt[3]{\frac{16 \cdot C}{\pi \cdot \tau_{allow}}}$$

$d_S = Shaft\ diameter\ [mm]$

$C = Torque\ [N \cdot mm]$

$\tau_{allow} = Allowableshearstress[N/mm^2]$

Misalignment Allowance

- $AngularMisalignment: (\theta \leq 1° - 2°)(typicalforjawcouplings)$.
- $ParallelMisalignment: (\Delta y \leq 0.25 - 0.5$,mm).
- $AxialMisalignment(EndFloat): (\Delta x \leq 1$,mm)

3.3. Concept Selection

Flexible Jaw Coupling (Elastomer Spider):

- Absorbs vibration and shock.
- Handles angular, parallel, and axial misalignment.
- Compact, simple, no lubrication required.

3.4. Concept Design Description

Component	Material	Function
Hub A	Cast Iron / Steel	Mounted on driving shaft, transmits torque
Hub B	Cast Iron / Steel	Mounted on driven shaft
Spider	Polyurethane / Rubber	Flexible element, absorbs vibration & misalignment

Working Principle:

Torque flows **Hub A → Spider → Hub B**. The elastomer deforms slightly, damping vibration and compensating misalignment. Under overload, it acts as a protective element.

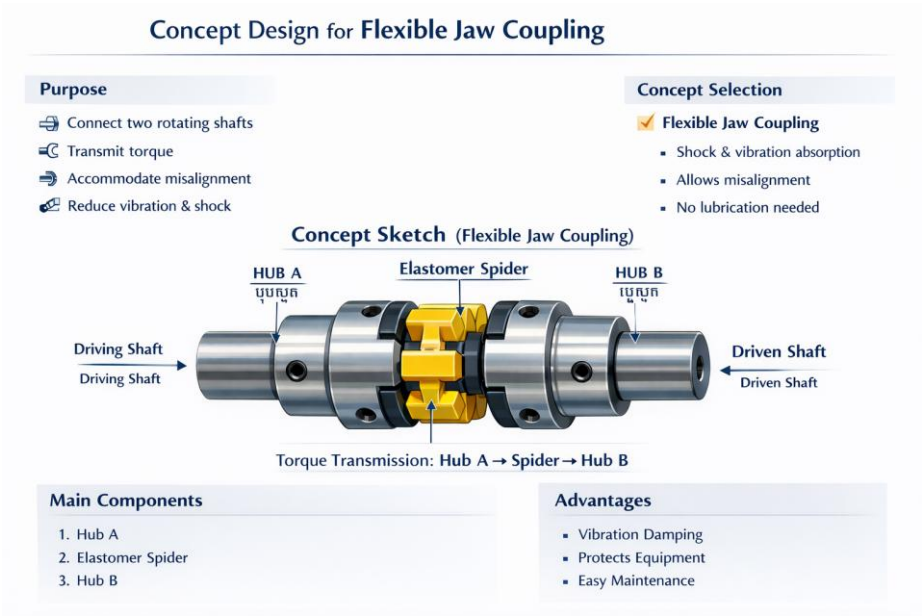


Figure 6: Hub A and Hub B

3.5. Design Considerations

- **Dynamic Balancing:** Required for speeds > 3600 rpm.
- **Spacer Option:** For large machines, allows maintenance without moving heavy motors.
- **Attachment Methods:** Keyway, split-hub, or taper lock depending on torque and speed.
- **Environment:** Material choice depends on dust, heat, or chemical exposure.

4. Maintenance

Maintenance of couplings is essential to ensure efficient power transmission, reduce vibration, prevent unexpected failures, and extend the service life of machinery. Proper and regular maintenance minimizes downtime and repair costs while improving overall system reliability.

- **Routine Inspection**

Regularly inspect the coupling for wear, cracks, corrosion, or deformation.
Check bolts, keys, and fasteners to ensure they are tight and secure. Look for abnormal noise, vibration, or temperature rise during operation.

- **Alignment Check**

Ensure proper **shaft alignment** (angular, parallel, and axial).
Misalignment can cause excessive stress on the coupling, bearings, and shafts, leading to premature failure. Alignment should be checked after installation and periodically during operation.

- **Lubrication**

For couplings that require lubrication (such as gear couplings or universal joints), apply the recommended lubricant. Use the correct type and amount of lubricant to reduce friction and wear. Replace old or contaminated lubricant regularly.

Cleaning

Keep the coupling free from dust, dirt, oil leaks, and debris. Contaminants can accelerate wear and reduce coupling performance.

- **Replacement of Worn Parts**

Replace worn or damaged components such as elastomer inserts, bushings, seals, or gear teeth promptly. Delayed replacement may cause damage to connected equipment.

- **Monitoring Operating Conditions**

Avoid operating beyond the coupling's rated torque, speed, and temperature limits.
Sudden overloads or frequent shock loads should be minimized.

- **Preventive Maintenance Schedule**

Establish a preventive maintenance plan based on operating hours and working conditions.
Record inspection and maintenance activities for future reference.

- **Safety Considerations**

Always stop the machine and isolate power before performing maintenance.
Use protective guards and follow standard safety procedures.
Conclusion (Maintenance)

Effective coupling maintenance ensures smooth operation, reduces vibration and energy loss, and prevents costly mechanical failures. Regular inspection, proper alignment, lubrication, and timely replacement of worn parts are key factors for reliable and long-term performance

5. The advantages and disadvantages

Advantages	Disadvantages
• Transmits power and torque between two shafts. • Connects driving and driven shafts efficiently. • Allows slight angular, parallel, and axial misalignment. • Reduces shock loads during starting and stopping. • Damps vibration and noise in machines. • Protects shafts and bearings from damage. • Acts as a safety device during overload (in some types). • Allows easy assembly and disassembly of machines. • Compensates for thermal expansion of shafts. • Improves service life of connected components. • Suitable for different speeds and load conditions. • Available in many types for different applications.	• Cannot tolerate large misalignment. • Excessive misalignment causes wear and failure. • Flexible elements wear out over time. • Some couplings require regular lubrication. • Maintenance and inspection are necessary. • High-quality couplings can be costly. • Flexible couplings may cause small power loss. • Some couplings are bulky and require more space. • Improper selection can lead to vibration problems. • Not suitable for very high shock loads in some cases.

Figure 7: The table of advantage and disadvantage

6. Application of Coupling

Couplings are essential components in mechanical engineering, used to connect two shafts together for the purpose of transmitting power. Their compact design, ease of installation, and ability to accommodate misalignment make them indispensable across a wide range of industries. Here's a deeper look into their applications:

- **Industrial Applications**

  - **Conveyor systems:** Couplings connect motor shafts to rollers or gearboxes, ensuring smooth material transport in factories, warehouses, and airports.
  - **Gearbox-to-gearbox connections:** Used in multi-stage gear systems to transfer torque while compensating for minor misalignments.
  - **Compressors and turbines:** In power plants and jet engines, couplings transmit high-speed rotational energy between components while absorbing vibration.

- **Household Appliances**

  - **Lawnmowers:** Couplings link the engine to the blade shaft, allowing for efficient torque transfer and easy maintenance.
  - **Washing machines and tumble dryers:** Flexible couplings connect motors to drums, reducing noise and wear while handling load variations.
  - **X-ray machines:** Precision couplings ensure accurate alignment between rotating components in medical imaging devices.

- **Automotive and Aerospace**

  - **Drive shafts and steering systems:** Couplings allow for angular movement and vibration damping in vehicles.
  - **Aircraft control systems:** Used in actuation mechanisms to ensure reliable transmission of control inputs under dynamic conditions.

For making this point more clearly, we have some examples in the real-world-life. These coupling are used in the tire-company by connecting with motor, which aide the rotation more smoothly and effectively.

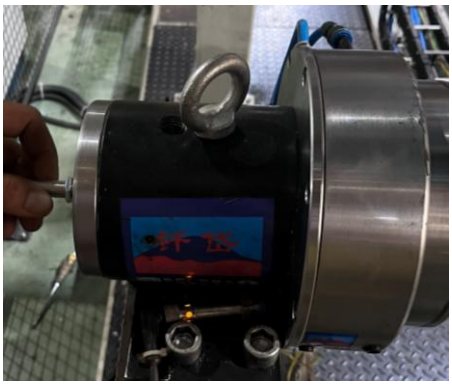


Figure 8: Pneumatic coupling

This coupling is a **pneumatic coupling** which connected to an electric motor and is mainly used in systems that require controlled rotation. It can be integrated with sensors and is controlled by airflow to open or close during operation. This type of coupling is commonly used in automated or precision systems. However, it is more susceptible to mechanical problems such as **tensile stress**, **bearing failure**, **shaft damage**, and **general wear due to misalignment**.

- **Tensile stress** (Figur2)**:**

  - When torque is transmitted through the coupling, the shaft and connected parts experience tensile and shear stresses.
  - If the airflow control causes abrupt engagement/disengagement, stress concentrations rise, leading to crack initiation.

- **Bearing failure** (Figure2)**:**

  - Bearings inside or near the coupling handle radial and axial loads.
  - Misalignment or uneven loading increases friction and heat, accelerating wear and reducing lubrication effectiveness.

- **General wear** (Figure3)**:**

  - Pneumatic couplings rely on frictional surfaces or clamping mechanisms.
  - Repeated engagement/disengagement under load causes surface wear.
  - Dust, vibration, or poor lubrication worsen degradation.

- **Shaft damage** (Figure4)**:**

  - Improper alignment between motor and driven shaft causes bending stresses.
  - Over time, this leads to fretting, scoring, or even fatigue cracks in the shaft.

Figure 11:Shaft damage

Figure 11: General wear

Figure 11: Bearing failure

Moreover, this coupling is **a universal coupling** used to transmit power from the gearbox to the rubber grinding machine. Its main function is to allow power transmission while accommodating angular misalignment between shafts. This coupling operates at high rotational speed and transmits very high torque, which is required for grinding expired rubber so that it can be reused. Due to the heavy load and high power, the coupling m ust be robust and well-maintained

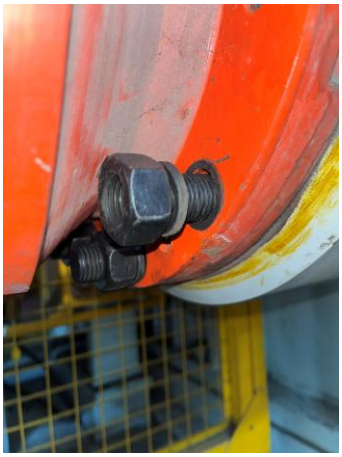


Figure 12:Universal coupling

IV. Result

This report on **Mechanical Couplings** demonstrates a solid understanding of the fundamental concepts of power transmission systems and reflects an appropriate academic level for a fourth-year mechanical engineering student.

- ## Technical Accuracy

The report correctly defines couplings and explains their primary functions, including torque transmission, misalignment compensation, vibration damping, and overload protection. The classification of couplings into **rigid, flexible, universal, and temporary couplings** is technically correct and well explained. The discussion of **misalignment types (angular, parallel, and axial)** shows good theoretical understanding and aligns with standard mechanical design principles.

The section on **materials** is technically sound, with correct justification for material selection based on strength, corrosion resistance, damping capability, and application environment. The explanation of **jaw couplings, gear couplings, universal joints, and fluid couplings** reflects correct mechanical behavior and industrial relevance.

- ## Engineering Application & Practical Insight

A strong point of this report is the inclusion of **real industrial applications**, such as conveyor systems, compressors, automotive drive shafts, and rubber grinding machines. The analysis of **pneumatic and universal couplings**, including failure modes (tensile stress, bearing failure, wear, shaft damage), demonstrates good practical awareness of mechanical stresses and operating conditions.

The **concept design section** is well structured and shows engineering thinking by considering:

- Design intent
- Operating conditions
- Material selection
- Advantages and limitations

This indicates the students understand not only *what* a coupling is, but *how and why* it is selected and designed.

- ## Structure and Organization

The report follows a logical structure, starting from theory and progressing to application, design, and maintenance. Headings and subheadings are clear, making the report easy to follow.

Figures and tables support the explanations, although some figures could benefit from clearer captions and references within the text.

V. Conclusion

This report demonstrates a solid foundation of technical knowledge, strong practical relevance, and an accurate application of mechanical engineering principles. The study effectively integrates theoretical concepts with practical considerations, showing how couplings can be selected, positioned, and maintained to achieve optimal performance. The inclusion of formulas and clear explanations of their usage enhances the technical rigor of the work, making it a valuable resource for both academic study and engineering practice.

With further development—particularly through deeper quantitative analysis, comparative case studies, and improved presentation of data—the report has the potential to reach an excellent academic standard. It already provides meaningful guidance on how to select the appropriate coupling based on operational conditions, environmental factors, and system requirements. This ensures that engineers and technicians can make informed decisions to maximize efficiency, reliability, and safety in mechanical systems.

Moreover, the report emphasizes the importance of long-term maintenance strategies, offering practical insights into inspection, lubrication, and preventive measures. These recommendations contribute to extending the service life of couplings, reducing downtime, and improving overall system performance. By combining theoretical understanding with practical application, the report serves not only as an academic document but also as a technical reference for real-world engineering challenges.

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